

INTELLIGENT DISASTER MANAGEMENT & RELOCATION PLATFORM

CENTRAL WATER COMMISSION (CWC) FLOOD FORECAST & HAZARD MITIGATION SYSTEM

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1. Executive System Overview & End-to-End Workflow

The **Intelligent Disaster Management & Relocation Platform** is an enterprise-grade emergency decision support system engineered for real-time hazard detection, automated ML risk classification, GIS spatial route planning, carrying capacity evaluation, and multi-channel emergency alert broadcasting. Designed in compliance with Central Water Commission (CWC) and National Disaster Management Authority (NDMA) guidelines, the platform bridges the gap between high-level command centers, field responders, and vulnerable citizens.

End-to-End Operational Pipeline Workflow:

- **Step 1: Sensor & Weather Data Ingestion:** Real-time rainfall telemetry (mm/24h), station hydrograph data, soil moisture indices, and digital elevation models (DEM) are ingested continuously into the backend PostGIS database.
- **Step 2: Machine Learning Threat Classification:** A Random Forest & Gradient Boosted Classifier evaluates environmental parameters (rainfall amount, intensity, terrain slope angle, soil erodibility K-factor, land-use) to compute threat scores (0.0 to 1.0) and categorize hazards into LOW, MODERATE, HIGH, or CRITICAL levels.
- **Step 3: GIS Red Zone Polygon Generation:** Spatial bounding boxes and high-risk hazard polygons are dynamically generated on Leaflet map layers, identifying unstable landslide slopes and inundated river flood plains.
- **Step 4: Relief Shelter Carrying Capacity Assessment:** The capacity engine evaluates nearby relief centers for available beds, water supply duration, food supply days, medical kit inventories, power backups, and sanitation units.
- **Step 5: Multi-Objective Relocation Optimization:** A mathematical linear programming solver assigns vulnerable citizen population groups to nearest safe relief shelters without exceeding maximum carrying capacities, prioritizing high-risk children and elderly.
- **Step 6: Multi-Channel Emergency Alert Dispatch:** High-priority emergency bulletins are broadcast instantly across Web Push Notifications, Emergency SMS, Email advisories, and live CWC warning tickers in 6 Indian languages.

2. Role-Based Person Dashboards

The platform provides tailored role-based authentication and specialized interfaces suited for different stakeholders during a disaster crisis:

A. Admin Command & Control Dashboard (NDMA / District Collectorate Chief)

Target Persona: District Collector, NDMA Command Officers, Disaster Management Authority Chiefs.

Key Responsibilities: High-level operational oversight, strategic resource allocation, emergency alert authorization, and system maintenance.

Dashboard Features:

- **Full Control Center Access:** Unrestricted access to all 11 modules including Red Zone Analysis, Relocation Solver, and System Health.
- **Emergency Alert Authorization:** Capability to compose, broadcast, acknowledge, and resolve high-severity emergency advisories.
- **Relocation Optimization Solver:** Execute multi-objective assignment algorithms for large-scale population evacuations.
- **Infrastructure Health Monitor:** Real-time diagnostic metrics on FastAPI application servers, spatial databases, and ML models.

- **Shelter Management:** Register new relief camps, modify maximum capacities, and override occupancy allocations.

B. Citizen Safety Portal (General Public / Evacuees)

Target Persona: Local residents, displaced citizens, tourists, and vulnerable community members in disaster sectors.

Key Responsibilities: Receiving localized hazard bulletins, locating nearest safe relief shelters, requesting emergency assistance, and navigating safe evacuation routes.

Dashboard Features:

- **Multi-Language Support (6 Languages):** Instant 1-click portal translation in English (EN), Malayalam (ML), Hindi (HI), Tamil (TA), Kannada (KN), and Telugu (TE).
- **Instant SOS Emergency Button:** Hot Pink/Magenta prominent SOS trigger sending high-priority GPS location coordinates to NDRF rescue teams.
- **CWC Live Rainfall Ticker:** Real-time warning ticker bar highlighting station hydrograph readings (e.g. 340.5mm rainfall, Kabini river level).
- **Nearest Safe Shelter Cards:** Displays available bed capacity, physical street address, emergency helpline numbers, and 1-click route navigation.
- **Interactive Evacuation Routing:** Calculates shortest path bypassing active red zones directly from user's current GPS location.

C. Field Operations & Tactical Response Dashboard (NDRF / First Responders)

Target Persona: NDRF Captains, Fire & Rescue Teams, Paramedics, and Relief Shelter Managers on the ground.

Key Responsibilities: Ground-level rescue operations, shelter occupancy check-ins, field hazard reporting, and offline emergency guidance.

Dashboard Features:

- **Tactical Map View:** Specialized GIS overlay showing active SOS signals, flooded road blockages, and landslide debris locations.
- **Shelter Check-In Management:** Real-time updating of shelter headcount as evacuees arrive at relief camps.
- **Offline Emergency Protocol:** Access pre-downloaded detour maps, survival guidelines, and emergency toll-free phone directories during cellular tower outages.
- **Field Hazard Reporting:** Submit ground truth observations directly to the Machine Learning risk assessment engine.

3. Detailed Navbar Feature Guide & Technical Implications

The navigation sidebar provides access to 11 modular features. Below is a comprehensive breakdown of each navbar item, its core capability, and what it implies technically for the system architecture:

Navbar Module	Description & Target Functionality	Technical Implication & Underpinning
1. Overview Dashboard	High-level executive summary displaying disaster risk distribution, affected population stats, available shelter capacity, and 24-hour station hydrograph rainfall trends.	Aggregates SQL queries across shelter and hazard tables. Uses Chart.js for rendering doughnut charts, stacked bar charts, and hydrograph line graphs.
2. Citizen Safety Portal	Simplified emergency dashboard for evacuees featuring multi-language selection (6 languages), live CWC warning ticker, instant SOS alert button, and helpline phone numbers.	Persists language preference in localStorage. Features 6-language translation dictionary (`translations.ts`). Dispatches SOS payload with GPS coordinates.
3. Live Hazard Map	Full-screen interactive GIS map with layer toggles for Red Hazard Zones, Emergency Relief Shelters, Evacuation Routes, and custom map tiles (Street, Dark, Satellite).	Uses Leaflet / React-Leaflet with OpenStreetMap and Esri tiles. Performs client-side map click event handling to update selected origin coordinates.
4. Red Zone Analysis	Machine Learning hazard prediction tool. Allows users to adjust rainfall (mm), slope angle (°), soil erodibility (K-factor), and land-use to compute threat scores.	Connects to FastAPI `/api/v1/predict-risk` endpoint running a trained Random Forest classifier. Returns threat probabilities for LOW, MODERATE, HIGH, and CRITICAL.

5. Shelters & Capacity	Comprehensive directory of emergency relief camps with real-time bed occupancy, physical addresses, contact phone numbers, and modal to register new shelters.	Queries `/api/v1/shelters` database. Calculates bed occupancy percentages and dynamically renders color-coded capacity indicators.
6. Carrying Capacity	Evaluates shelter suitability based on resource availability (water supply days, food supply days, medical kits, power backup, sanitation units). Includes group evaluator.	Uses multi-criteria weighted scoring algorithm. Evaluates feasibility of assigning specified group sizes without exceeding shelter capacities.
7. Evacuation Planning	Calculates shortest safe evacuation route from any user-selected danger location to the nearest safe relief shelter, avoiding red hazard zones.	Implements Haversine distance formula and Dijkstra spatial routing graph algorithm. Integrates HTML5 Geolocation API for live GPS position fetching.
8. Relocation Plan	Intelligent multi-objective population relocation optimization engine. Allocates multiple population groups from hazard red zones to safe shelters.	Powered by linear optimization solver (`/api/v1/relocation/plan`). Prioritizes vulnerable citizens (children/elderly) and outputs transparent decision rationales.
9. Disaster Alerts	Multi-channel emergency broadcast operations dashboard for creating, filtering, acknowledging, and resolving high-severity disaster alerts.	Triggers multi-channel notification dispatches (Web Push, SMS, Email). Implements severity filtering (CRITICAL, HIGH, WARNING, INFO) and status lifecycle state.
10. Offline Emergency	Offline PWA disaster guide for network blackout scenarios. Displays pre-downloaded detour maps, shelter addresses, helpline phone directory, and survival guides.	Uses Progressive Web App (PWA) Service Worker (`sw.js`) and `localStorage` caching. Outlines native Bluetooth Mesh / Wi-Fi Direct peer-to-peer specs.
11. System Health	Real-time component diagnostic monitor showing live health status of Application Server, Spatial Database, ML Risk Classifier, and Routing Engine.	Queries `/health` endpoint. Returns JSON status of database connection, active port bindings, and model runtime readiness.

4. Summary Role Access Matrix

Navbar Module	Citizen (User)	Field Ops (Responder)	Admin (Control Center)
Overview Dashboard	—	✓ Full Access	✓ Full Access
Citizen Safety Portal	✓ Primary View	✓ Full Access	✓ Full Access
Live Hazard Map	✓ View Only	✓ Interactive	✓ Full Access
Red Zone Analysis	—	✓ Interactive	✓ Full Access
Shelters & Capacity	✓ View Only	✓ Update Occupancy	✓ Full Management
Carrying Capacity	—	✓ View Ratings	✓ Full Evaluator
Evacuation Planning	✓ Find My Route	✓ Tactical Routes	✓ Full Planning
Relocation Plan	—	✓ View Plan	✓ Run Solver
Disaster Alerts	✓ Receive Alerts	✓ View Advisories	✓ Broadcast & Resolve
Offline Emergency	✓ Offline Guide	✓ Offline Guide	✓ Offline Guide
System Health	—	—	✓ Diagnostic Admin